SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5117
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage**: 40%**

QUESTIONS: **Total Marks: 50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I N. Saiyashaswini 192425250 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Nyashasy

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis

Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

2. Block cipher Mode evaluation

Given:

→ plaintext blocks: P = [1010, 1010, 1111, 1010]

→ Key: K = 1100

→ Initialization Vector (IV): 0000

Assumption: use XOR with the key as a simplified block encryption function $E_K(K) = X \oplus K$.

1.1 Encryption using ECB mode:

In ECB mode, each plaintext block is XORed independently with the key

$$C_i = P_i \oplus K$$

Block	Plaintext	Key	ciphertext
1	1010	1100	0110
2	1010	1100	0110
3	1111	1100	0011
4	1010	1100	0110

ECB ciphertext: 0110 0110 0011 0110

1.2 Encryption using CBC mode:

In CBC mode, each plaintext block is XORed with the Previous ciphertext block before encryption

$$C_1 = E_K(P_1 \oplus IV)$$

$$C_i = E_K(P_i \oplus C_{i-1})$$

Block	Calculation	Ciphertext
1	1010 ⊕ 0000 ⊕ 1100	0110
2	1010 ⊕ 0110 ⊕ 1100	0000
3	1100 ⊕ 0000 ⊕ 1100	0011
4	1010 ⊕ 0011 ⊕ 1100	0101

CBC ciphertext : 0110 0000 0011 0101

1.3 compare the ciphertext outputs

ECB mode	CBC mode
0110 0110 0011 0110	0110 0000 0011 0101
Identical plaintext blocks produce identical ciphertext blocks	Identical plaintext blocks can produce different ciphertext bl
Patterns are more visible	patterns are better concealed.

1.4 Analysis :

CBC mode hides patterns betterns than ECB mod because each block depends on the previous ciphertc block. ECB encrypts identical plaintext blocks into

identical ciphertext blocks, revealing patterns. CBC reduces this repetition by chaining blocks together

Note: XOR with the key is a simplified demonstration, not a secure block cipher. Real CBC encryption uses a proper block cipher, and its IV must be unpredictable

5. ECC vs RSA comparison:

Given:

→ RSA key size = 1024 bits
→ ECC key size = 160 bits (equivalent security as specified)
→ Device = IOT device with limited processing capacity

2.1 computational cost

Factor	RSA	ECC
Key size	1024 bits	160 bits
Computation	Large-number modular exponentiation	Elliptic curve point operations
Processing Requirement	Generally higher	Generally lower
Memory usage	More memory for keys and operations	Less memory for keys and operations
Power consumption	Generally higher	Generally lower

2.2 which consumes less power and memory?

ECC generally consumes less power and memory because it provides comparable security with much smaller keys, reducing storage, communication overhead and many computational costs

2.3 which cryptosystem is better for IOT devices?

ECC is the better choice for the given IOT scenario

Reasons:

1. smaller keys
2. Lower power consumption
3. Less communication overhead
4. Efficient Processing
5. Strong security per key bit

Conclusion:

ECC is generally preferred for resource-constrained IOT devices because it offers strong security with smaller keys, lower memory requirements and often lower energy consumption